Day 166: Fairness: Demographic Parity, Equalized Odds, and Why They Conflict

MD Mutasim Billah | 52-Week Data Science to ML/AI Roadmap | Week 24, Day 6 | Complete study guide

Watch: Fairness, part 1 - Moritz Hardt - MLSS 2020, Tuebingen (virtual mlss2020) | FAT* 2018 Translation Tutorial: 21 Definitions of Fairness and Their Politics (ACM FAccT)

Script: `day166_fairness.py` · every number in this guide is copied by program from `day166_run_output.txt`, the real output of that script.

A model can be accurate overall and still treat groups differently. Today measures that for two UCI Adult models, by sex. Logistic regression selected 0.0790 of women and 0.2652 of men as high earners. Part of that follows the data (the base rates differ), part does not: among people who really were high earners, women were found less often (0.4956 against 0.6110). Deleting the sex column changed almost nothing, because the other columns reveal sex with AUC 0.9341. An identity shows why the usual criteria cannot all hold at once, and one threshold per group, chosen on half the rows, shows what each criterion costs.

0. Where today fits

DEFINITION — start here, no prior knowledge assumed

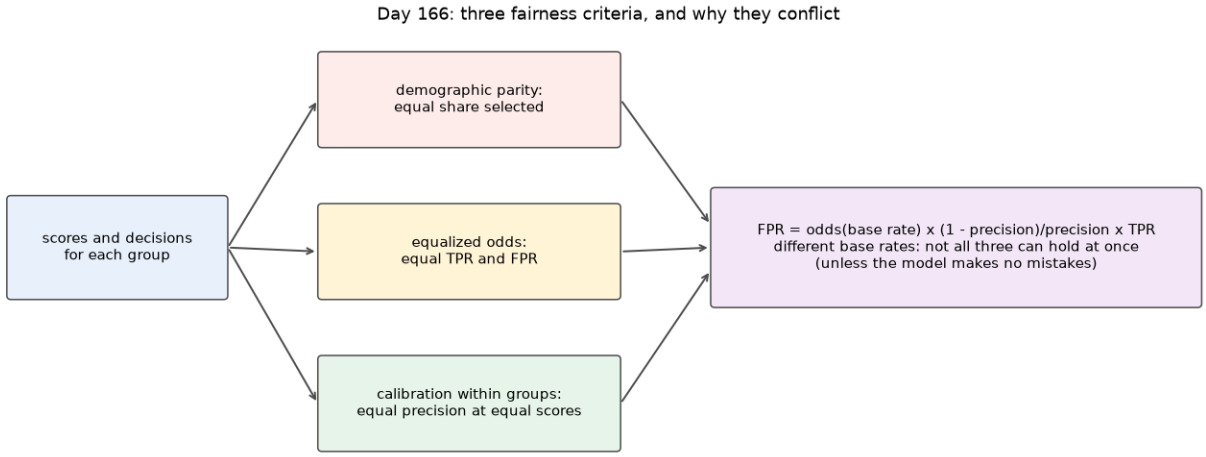
A model's **decisions** are compared across **groups** (here women and men). Four rates per group: the **selection rate** (share predicted positive), the **true positive rate** TPR (share of the truly positive who are selected), the **false positive rate** FPR (share of the truly negative who are selected) and **precision** (share of the selected who are truly positive).

Three common criteria: **demographic parity** asks for equal selection rates; **equalized odds** asks for equal TPR and equal FPR (**equal opportunity** asks only for equal TPR); **calibration within groups** asks that a score of 70% means 70% in every group, which makes precision at a given score equal.

REAL-WORLD SCENARIO

A lender approves loans with a credit model. Regulators ask: are women approved as often as men? Among people who would repay, are women approved as often? When the model says 'low risk', is it equally right for both? These are three different questions, and the answers can disagree.

SYSTEM DIAGRAM



One set of scores, three criteria; the identity on the right ties them to each group's base rate, so when base rates differ an imperfect model cannot meet all three.

The data and split are Week 24's: UCI Adult's training file, 24,130 fitting rows and 6,032 validation rows; the test file is kept for Day 167. The target is whether a person earned above $50K. Two models: Day 161's **logistic regression** and Day 109's **gradient boosting** (200 trees of depth 3). Decisions are 'probability above 0.5' unless a section says otherwise. Adult records sex as Female or Male; the lesson uses those two values as recorded.

Today's part for the week's build

`day166_exercise.py` is today's toolkit (rates per group, the gap between groups, a threshold for a chosen selection rate or TPR, per-group decisions, the FPR identity, and which feature each one-hot column came from), each checked by hand on two groups of four people. It is the sixth part of Day 167's report card.

ACTUAL OUTPUT

```
women: outcomes 1, 1, 0, 0, decisions 1, 0, 1, 0 -> by hand base 0.5, selected 0.5, TPR 1/2, FPR
    1/2, precision 1/2
  got [0.5, 0.5, 0.5, 0.5, 0.5]
men: outcomes 1, 0, 0, 0, decisions 1, 1, 0, 0 -> by hand base 0.25, selected 0.5, TPR 1, FPR 1/3,
    precision 1/2
  got [0.25, 0.5, 1.0, 0.3333, 0.5]
gap men - women: TPR by hand 1 - 0.5 = 0.5, got 0.5000; FPR 1/3 - 1/2 = -0.1667, got -0.1667
fpr_from_identity for the men: odds 0.25/0.75 = 1/3, x (1 - 0.5)/0.5 = 1, x TPR 1 = 0.3333; got
    0.3333
threshold_for_rate(0.1 to 1.0, 30%): the 70th percentile, by hand 0.7 + 0.3 x 0.1 = 0.73; got
    0.73, selecting 3 of 10
threshold_for_tpr(TPR 2/3): positives 0.2, 0.6, 0.8; the 1/3 quantile, by hand 0.2 + (2/3) x 0.4 =
    0.4667; got 0.4667, TPR 0.6667
decide with women at 0.2 and men at 0.5, scores 0.3, 0.6 (women) and 0.3, 0.6 (men): by hand 1, 1,
    0, 1; got [1, 1, 0, 1]
feature_of_column: 87 columns (Day 161's matrix has 87); sex has 2, relationship 6, marital-status
    7
```

How to read each section

Step	What it gives you
DEFINITION	What the idea is, in plain words. Start here.
REAL-WORLD SCENARIO	An everyday situation where the same idea shows up.
MATHEMATICS	The formula, when the section introduces one.
SPECIFICATION	The exact setup of the experiment.
IMPLEMENTATION	The real code from the script.
ACTUAL OUTPUT	What the script really printed, copied unchanged.
What it means	How to read the numbers, and what they do and do not show.
Why choose this? / Why NOT this?	When the idea is the right tool, and what it costs.

1. One model, two groups: the four rates

DEFINITION — start here, no prior knowledge assumed

Each model's decisions at 0.5 are split by sex, and the four rates are computed per group. Each gap (men minus women) gets a 95% bootstrap interval from Day 163 (1,000 resamples of the validation rows).

A control: the sex labels are shuffled 200 times, keeping the group sizes, and the same gap is measured between these random groups. Its spread shows how big a gap chance alone produces.

REAL-WORLD SCENARIO

A hiring screen passes 27% of male applicants and 8% of female applicants. Before calling that unfair, check how many of each were qualified, and whether the screen finds qualified people equally well.

MATHEMATICS

$$selection = P(d = 1 \mid group) \quad TPR = P(d = 1 \mid y = 1, group) \quad FPR = P(d = 1 \mid y = 0, group)$$

$$precision = P(y = 1 \mid d = 1, group) \quad \text{80\% rule: } selection_{women} / selection_{men} \geq 0.8$$

IMPLEMENTATION

```
def group_rates(y, d, groups):
    out = {}
    for g in np.unique(groups):
        yy, dd = y[groups == g], d[groups == g]
        out[g] = {"n": len(yy), "base": yy.mean(),
                  "sel": dd.mean(),
                  "tpr": dd[yy == 1].mean() if (yy == 1).any() else np.nan,
                  "fpr": dd[yy == 0].mean() if (yy == 0).any() else np.nan,
                  "ppv": yy[dd].mean() if dd.any() else np.nan}
    return out
def gap(y, d, groups, metric, a="Male", b="Female"):
    r = group_rates(y, d, groups)
    return r[a][metric] - r[b][metric]
```

ACTUAL OUTPUT

```
  Female: 1925 validation people, 0.1174 earn above $50K
  Male  : 4107 validation people, 0.3104 earn above $50K
  logistic: accuracy 0.8408
    Female: selected 0.0790, TPR 0.4956, FPR 0.0235, precision 0.7368
    Male  : selected 0.2652, TPR 0.6110, FPR 0.1095, precision 0.7153
    women's selection rate / men's = 0.298 (the '80% rule' asks for at least 0.8)
    gap men - women in sel: +0.1862 [+0.1684, +0.2033]; between random groups: spread 0.0111
    gap men - women in tpr: +0.1154 [+0.0455, +0.1833]; between random groups: spread 0.0264
    gap men - women in fpr: +0.0859 [+0.0731, +0.1002]; between random groups: spread 0.0088
    gap men - women in ppv: -0.0215 [-0.0931, +0.0527]; between random groups: spread 0.0275
  boosting: accuracy 0.8662
    Female: selected 0.0831, TPR 0.5796, FPR 0.0171, precision 0.8187
    Male  : selected 0.2600, TPR 0.6510, FPR 0.0840, precision 0.7772
    women's selection rate / men's = 0.320 (the '80% rule' asks for at least 0.8)
    gap men - women in sel: +0.1769 [+0.1588, +0.1943]; between random groups: spread 0.0111
    gap men - women in tpr: +0.0713 [+0.0057, +0.1396]; between random groups: spread 0.0254
    gap men - women in fpr: +0.0670 [+0.0553, +0.0790]; between random groups: spread 0.0086
    gap men - women in ppv: -0.0416 [-0.1062, +0.0213]; between random groups: spread 0.0281
Actual output from this lesson's run.
```

What it means

The base rates differ a lot: 0.1174 of the women and 0.3104 of the men earned above $50K. Logistic regression selected 0.0790 and 0.2652, a ratio of 0.298, far below the 80% rule's 0.8. The selection gap, +0.1862 [+0.1684, +0.2033], is many times the spread between random groups (0.0111): it is not chance.

The other rates tell a different story from selection. Among the truly positive, women were found less often (TPR gap +0.1154 [+0.0455, +0.1833]); among the truly negative, men were wrongly selected more often (FPR 0.0235 for women, 0.1095 for men). Precision was about equal: its gap -0.0215 has an interval [-0.0931, +0.0527] that includes zero.

Boosting was more accurate (0.8662 against 0.8408) and narrowed the TPR gap to +0.0713 [+0.0057, +0.1396], but its selection ratio was still 0.320.

WHY CHOOSE THIS?

Four rates per group show which kind of difference there is: who is selected, who is missed, who is wrongly picked.

WHY NOT THIS? (TRADEOFFS)

Each rate answers a different question; a model can look fair on one and unfair on another, as here.

2. 'Fairness through unawareness': delete the column

DEFINITION — start here, no prior knowledge assumed

The simplest proposal: do not show the model the sex column. Logistic regression is retrained without it, then also without *relationship* (whose values include Husband and Wife) and *marital-status*.

To see why this might not work, a second logistic regression is trained to predict sex from every column the model still has. If it can, the model can still use sex indirectly; the columns that carry it are **proxies**.

REAL-WORLD SCENARIO

A bank stops recording applicants' race but keeps their postcode. In a segregated city, the postcode tells the model nearly as much as the deleted column did.

IMPLEMENTATION

```
for drop in ([], ["sex"], ["sex", "relationship"], ["sex", "relationship", "marital-status"]):
    cols = ~np.isin(feat, drop)
    w, _ = d161.fit_logistic(Xo[fi][:, cols], y[fi]); p = d161.predict_logistic(w, Xo[va][:,
        cols])
    c2 = cols & (feat != "sex")
    w2, _ = d161.fit_logistic(Xo[fi][:, c2], s[fi])
    auc_s = d117.roc_auc(s[va], d161.predict_logistic(w2, Xo[va][:, c2]))
```

ACTUAL OUTPUT

```
  removed nothing                          : accuracy 0.8408, AUC 0.8972; sex predictable from the rest with AUC 0.9341
    selected women 0.0790, men 0.2652; TPR women 0.4956, men 0.6110; FPR women 0.0235, men 0.1095
  removed sex                              : accuracy 0.8400, AUC 0.8956; sex predictable from the rest with AUC 0.9341
    selected women 0.0842, men 0.2605; TPR women 0.5221, men 0.6008; FPR women 0.0259, men 0.1073
  removed sex, relationship                : accuracy 0.8397, AUC 0.8936; sex predictable from the rest with AUC 0.8637
    selected women 0.0691, men 0.2642; TPR women 0.4513, men 0.6071; FPR women 0.0182, men 0.1098
  removed sex, relationship, marital-status    : accuracy 0.8037, AUC 0.8372; sex predictable from the rest with AUC 0.8010
    selected women 0.0862, men 0.1838; TPR women 0.3274, men 0.4275; FPR women 0.0541, men 0.0742
Actual output from this lesson's run.
```

What it means

Removing sex alone barely moved anything: selection 0.0842 for women and 0.2605 for men, accuracy 0.8400. The remaining columns predict sex with AUC 0.9341, so the information was still there.

Removing relationship as well lowered that AUC to 0.8637, and women's selection fell to 0.0691, further from men's. Removing marital-status too narrowed the selection gap (women 0.0862, men 0.1838) but at a real cost: accuracy fell to 0.8037, and both groups' TPR fell (women 0.3274, men 0.4275) while women's FPR rose to 0.0541. Sex was still predictable with AUC 0.8010. The gap shrank while the model got worse for everyone, and it did not become equal in any of the senses above.

WHY CHOOSE THIS?

Simple, and sometimes legally required: the protected column is not an input.

WHY NOT THIS? (TRADEOFFS)

Proxies carry the same information; deleting them costs accuracy, and the resulting rates can move either way.

3. Why the criteria conflict

DEFINITION — start here, no prior knowledge assumed

For any group, four of its numbers are tied together by an identity that follows from the definitions alone. Write the group's base rate as odds, base / (1 – base). Then FPR is fixed by the odds, the precision and the TPR.

Within-group calibration means precision depends on the score, not the group. If two groups have equal precision and equal TPR but different base rates, the identity forces their FPRs apart by exactly the ratio of their odds. So unless the model is perfect, calibration and equalized odds cannot both hold.

REAL-WORLD SCENARIO

Two clinics use the same well-calibrated test. In one clinic the disease is three times as common (in odds). If the test catches the same share of sick patients in both and is equally reliable when positive, it must raise false alarms on healthy patients about three times as often in the high-rate clinic.

MATHEMATICS

$$FPR = base / (1 - base) \cdot (1 - precision) / precision \cdot TPR$$

$$\text{equal precision and } TPR \Rightarrow FPR_{men} / FPR_{women} = odds_{men} / odds_{women}$$

IMPLEMENTATION

```
def fpr_from_identity(base, tpr, ppv):
    return base / (1 - base) * (1 - ppv) / ppv * tpr
```

ACTUAL OUTPUT

```
  logistic:
    Female: mean probability 0.1173 against a rate of 0.1174 (ECE 0.0208); precision 0.7368, TPR 0.4956, FPR 0.0235;
        identity gives FPR 0.0235
    Male  : mean probability 0.3126 against a rate of 0.3104 (ECE 0.0178); precision 0.7153, TPR 0.6110, FPR 0.1095;
        identity gives FPR 0.1095
  boosting:
    Female: mean probability 0.1167 against a rate of 0.1174 (ECE 0.0128); precision 0.8187, TPR 0.5796, FPR 0.0171;
        identity gives FPR 0.0171
    Male  : mean probability 0.3093 against a rate of 0.3104 (ECE 0.0101); precision 0.7772, TPR 0.6510, FPR 0.0840;
        identity gives FPR 0.0840
  base-rate odds: women 0.1330, men 0.4502 (ratio 3.38); with equal precision and equal TPR the identity would force men's
      FPR to be 3.38 times women's
Actual output from this lesson's run.
```

What it means

The identity matched the measured FPR exactly for both models and both groups (for example 0.0235 and 0.1095 for logistic regression). Both models were close to calibrated within each sex: logistic regression's mean probability for women was 0.1173 against a rate of 0.1174, with ECE 0.0208 (Day 165 measured its chance floor at about 0.0135).

The odds were 0.1330 for women and 0.4502 for men, a ratio of 3.38. That is why a calibrated model here selects fewer women and wrongly selects more men: the base rates differ, and calibration passes that difference through.

WHY CHOOSE THIS?

One line of algebra explains why the criteria disagree, before any argument about which one is right.

WHY NOT THIS? (TRADEOFFS)

It says that a choice must be made, not which one; that depends on what each kind of error costs and to whom.

4. One threshold per group, chosen on half the rows

DEFINITION — start here, no prior knowledge assumed

If one threshold gives unequal rates, each group can get its own. For **parity**, each group's threshold is set so that it is selected at the overall rate of the single-threshold model; for **equal opportunity**, so that its TPR equals the overall TPR. Both use boosting.

The thresholds are chosen on a random half of the validation rows (A) and everything is measured on the other half (B), 50 times with different halvings. A rule that only works on the rows it was tuned on would show it here.

REAL-WORLD SCENARIO

A university admits by test score. A separate cut-off per school district can equalise admission rates, or equalise the admission rate of students who would graduate; the two cut-offs are different.

IMPLEMENTATION

```
def threshold_for_rate(p, rate):
    return np.quantile(p, 1 - rate)
def threshold_for_tpr(p, y, tpr):
    return np.quantile(p[y == 1], 1 - tpr)
def decide(p, groups, thr):
    d = np.zeros(len(p), bool)
    for g, t in thr.items():
        d[groups == g] = p[groups == g] > t
    return d
```

ACTUAL OUTPUT

```
means over 50 random halvings (boosting; thresholds chosen on half A, everything below measured on half B):
  one threshold 0.5: thresholds women 0.500, men 0.500; accuracy 0.8669 (spread 0.0046); change against one threshold on the
      same halves +0.0000 (spread 0.0000)
    women: selected 0.084, TPR 0.582, FPR 0.017, precision 0.826
    men:   selected 0.261, TPR 0.653, FPR 0.084, precision 0.779
  parity (equal selection): thresholds women 0.124, men 0.624; accuracy 0.8499 (spread 0.0051); change against one threshold
      on the same halves -0.0169 (spread 0.0052)
    women: selected 0.206, TPR 0.899, FPR 0.112, precision 0.522
    men:   selected 0.205, TPR 0.559, FPR 0.046, precision 0.847
  equal opportunity (equal TPR): thresholds women 0.437, men 0.515; accuracy 0.8673 (spread 0.0047); change against one
      threshold on the same halves +0.0004 (spread 0.0010)
    women: selected 0.097, TPR 0.641, FPR 0.023, precision 0.793
    men:   selected 0.254, TPR 0.643, FPR 0.079, precision 0.786
Actual output from this lesson's run.
```

What it means

Parity worked on half B (women 0.206, men 0.205 selected) by moving women's threshold down to 0.124 and men's up to 0.624. The cost: accuracy changed by -0.0169 (spread 0.0052 over the 50 halvings), women's precision fell to 0.522, and the FPR gap reversed (women 0.112, men 0.046). Women's TPR rose to 0.899 while men's fell to 0.559.

Equal opportunity needed much smaller moves (0.437 and 0.515) and equalised TPR on half B (0.641 and 0.643). Precision stayed about equal (0.793 and 0.786), so, as section 3 predicts, the FPRs stayed apart: 0.023 for women and 0.079 for men, more than three times as high.

A first comparison of the accuracies alone (0.8673 against 0.8669) would have suggested equal opportunity was free or even helped. That difference is far inside the spread of either (0.0047), so the change was measured on the same halvings instead: +0.0004 with spread 0.0010, no measurable cost in either direction.

WHY CHOOSE THIS?

Per-group thresholds reach the chosen criterion exactly, without retraining, and the cost can be measured.

WHY NOT THIS? (TRADEOFFS)

They need the group at decision time (often not allowed), fix one criterion while moving the others, and parity here cost accuracy and women's precision.

5. Small groups: race × sex

DEFINITION — start here, no prior knowledge assumed

Fairness is often checked for more than one attribute. Crossing Adult's five race values with sex gives ten groups; some are tiny. Each group's selection rate gets a 95% interval.

The first draft used the bootstrap. For a group where nobody was selected, every resample also has nobody selected, so the bootstrap interval collapses to [0, 0], claiming a certainty the data cannot give. The final version uses Day 163's **Wilson interval**, which stays honest at 0 out of n; the bootstrap is printed beside it for comparison.

REAL-WORLD SCENARIO

A pilot programme approves none of the 16 applicants from one small community. That does not prove the rate is zero; it is consistent with up to about one in five.

MATHEMATICS

$$\text{Wilson: } \left(r + z^2/2n \pm z \sqrt{\left(r(1-r)/n + z^2/4n^2 \right)} \right) / (1 + z^2/n), \quad r = \text{selected} / n, \quad z = 1.96$$

IMPLEMENTATION

```
lo, hi = d163.wilson_ci(d[m].sum(), n); lo = max(lo, 0.0)
reps = d163.bootstrap_reps(np.mean, (d[m].astype(float),), B, seed=4)
blo, bhi = d163.percentile_ci(reps)
```

ACTUAL OUTPUT

```
  Amer-Indian-Eskimo Female:   23 people, base rate 0.000, selected 0.043 Wilson [0.008, 0.210] (bootstrap [0.000, 0.130]),
      TPR undefined (nobody positive)
  Amer-Indian-Eskimo Male  :   36 people, base rate 0.111, selected 0.111 Wilson [0.044, 0.253] (bootstrap [0.028, 0.222]),
      TPR 0.750 (of 4)
  Asian-Pac-Islander Female:   67 people, base rate 0.194, selected 0.104 Wilson [0.052, 0.200] (bootstrap [0.045, 0.179]),
      TPR 0.462 (of 13)
  Asian-Pac-Islander Male  :  115 people, base rate 0.304, selected 0.339 Wilson [0.259, 0.430] (bootstrap [0.252, 0.435]),
      TPR 0.657 (of 35)
  Black              Female:  280 people, base rate 0.075, selected 0.043 Wilson [0.025, 0.073] (bootstrap [0.021, 0.068]),
      TPR 0.476 (of 21)
  Black              Male  :  295 people, base rate 0.220, selected 0.139 Wilson [0.104, 0.183] (bootstrap [0.102, 0.180]),
      TPR 0.538 (of 65)
  Other              Female:   16 people, base rate 0.000, selected 0.000 Wilson [0.000, 0.194] (bootstrap [0.000, 0.000]),
      TPR undefined (nobody positive)
  Other              Male  :   33 people, base rate 0.091, selected 0.091 Wilson [0.031, 0.236] (bootstrap [0.000, 0.212]),
      TPR 0.667 (of 3)
  White              Female: 1539 people, base rate 0.125, selected 0.091 Wilson [0.078, 0.106] (bootstrap [0.077, 0.107]),
      TPR 0.599 (of 192)
  White              Male  : 3628 people, base rate 0.322, selected 0.270 Wilson [0.256, 0.285] (bootstrap [0.256, 0.284]),
      TPR 0.657 (of 1168)
  10 groups x 4 rates = 40 comparisons: Day 164's multiple-comparisons problem returns
Actual output from this lesson's run.
```

What it means

The Other Female group had 16 people and nobody selected: Wilson [0.000, 0.194], bootstrap [0.000, 0.000]. White men, the largest group (3628 people), had an interval only 0.256 to 0.285 wide. In the Amer-Indian-Eskimo Female group (23 people) nobody was a high earner, so TPR is undefined and the one person selected was a false positive.

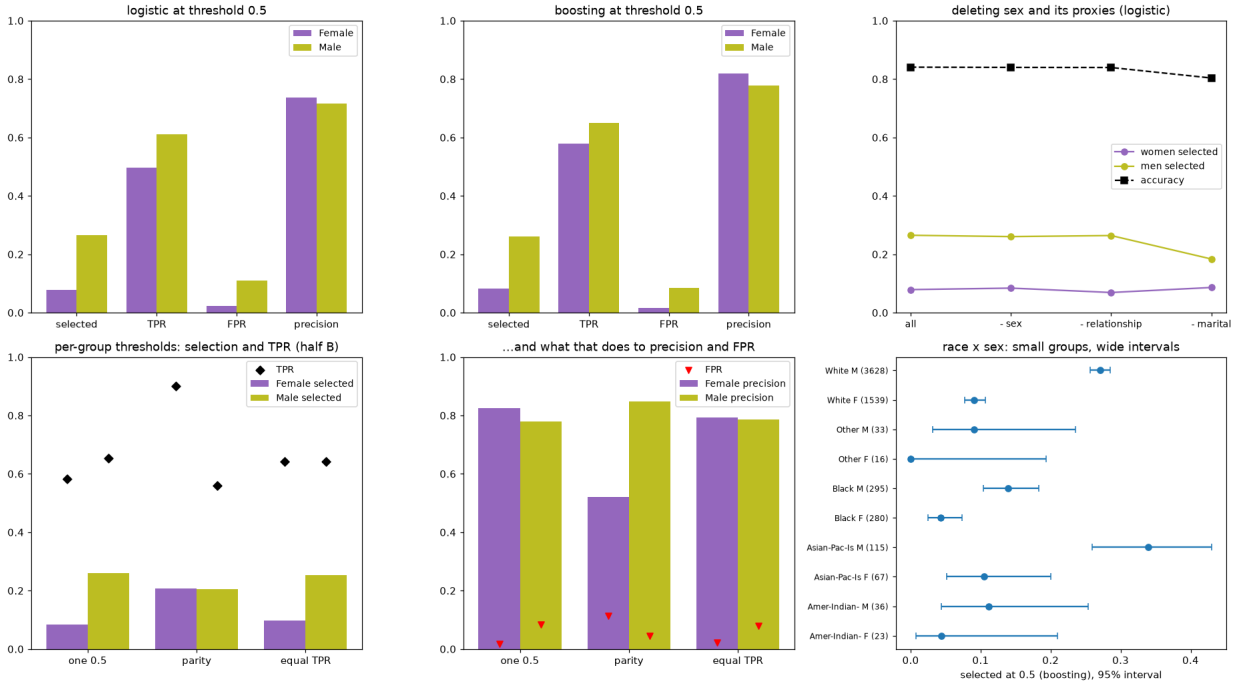
Ten groups and four rates make 40 comparisons, so some will look different by chance: Day 164's multiple-comparisons problem returns. Small groups need wide intervals and a correction, or more data, before any conclusion.

WHY CHOOSE THIS?

Checking crossed groups finds problems that one attribute at a time hides.

WHY NOT THIS? (TRADEOFFS)

Groups get small fast; intervals become too wide to conclude much, and the number of comparisons grows.



Top: the four rates per sex for both models at 0.5; selection and accuracy as sex and its proxies are deleted. Bottom: selection, TPR, precision and FPR on half B under the three threshold rules; selection rate with its Wilson interval for each race × sex group.

Interview preparation — technical questions

Q: Your model selects 8% of women and 27% of men. Is it unfair?

A: Not by that alone: the base rates were 12% and 31%. Check TPR (0.50 against 0.61 here, a real gap) and precision (about equal).

Q: Does removing the protected column make a model fair?

A: Rarely: the other Adult columns predicted sex with AUC 0.93, and removing proxies too cost accuracy (0.84 to 0.80) without equalising the rates.

Q: How do you tell a real gap from noise?

A: A bootstrap interval for the gap, and the gap between random groups of the same sizes as a control (spread 0.011 against a gap of 0.186).

Q: How would you make TPR equal across groups?

A: A threshold per group chosen on held-out rows; here it cost no measurable accuracy (+0.0004, spread 0.0010), but FPR stayed unequal.

Q: A group of 16 has nobody selected. What interval do you report?

A: Wilson, [0, 0.19]; the bootstrap collapses to [0, 0].

Interview preparation — theoretical questions

Q: Why can't calibration and equalized odds both hold?

A: FPR = odds(base) × (1 – precision)/precision × TPR; with equal precision and TPR, different base rates force different FPRs.

Q: Demographic parity or equal opportunity?

A: Parity ignores who is truly positive and here cost accuracy and women's precision; equal opportunity equalises who is found. The choice depends on which error harms whom.

Q: When is the conflict avoidable?

A: When base rates are equal, or the model is perfect.

Q: Why measure the threshold rules on a different half?

A: A threshold tuned on the same rows reaches its target there by construction; half B shows whether it holds.

Key takeaway and what's next

> **KEY TAKEAWAY**
> Fairness is several questions, not one. Measure all four rates per group with intervals and a random-group control; do not expect deleting a column to help; know that with different base rates calibration and equal error rates conflict; and if you set a threshold per group, choose it on one half and measure it on another.

- Logistic: selected women 0.0790, men 0.2652; TPR gap +0.1154.
- Sex predictable from the other columns: AUC 0.9341.
- Odds ratio 3.38: equal precision and TPR force unequal FPR.
- Parity cost -0.0169 accuracy; equal TPR +0.0004 (spread 0.0010).

Suggested watch order, if starting from zero

- 1. Fairness, part 1 - Moritz Hardt - MLSS 2020, Tuebingen (virtual mlss2020) — the criteria and why they conflict, by one of the authors of equal opportunity.
- 2. FAT* 2018 Translation Tutorial: 21 Definitions of Fairness and Their Politics (ACM FAccT) — the many definitions of fairness and what each assumes.

Day 167 preview

- **The report card.** SHAP and LIME built from scratch, then all six of the week's tools applied to one model and scored once on Adult's untouched test file.